An Ethical Domain-Specific Generative AI Framework for Adolescent Mental Health Triage in India

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Abstract

Adolescent mental health disorders have rapidly escalated into a profound public health crisis within India, characterised by rising prevalence rates of psychological distress and an overwhelming treatment gap currently estimated at nearly 95%.[1] Concurrently, the global proliferation of generative Artificial Intelligence (AI) and Large Language Models (LLMs) has catalysed the deployment of digital mental health tools designed to bridge this accessibility void.[4] However, the foundational architecture of contemporary AI-driven wellness systems predominantly prioritises continuous user engagement and therapeutic simulation. This pattern introduces severe ethical vulnerabilities into the clinical ecosystem, including the systemic mismanagement of acute psychiatric crises, the cultivation of maladaptive emotional dependencies through deceptive empathy, and a profound lack of clinical accountability.[1]

This comprehensive research report details the conceptualisation and architectural design of a platform-neutral, domain-specific generative AI framework engineered exclusively for structured mental health triage among Indian adolescents aged 16 to 18. Distinct from engagement-driven conversational agents, the proposed framework integrates rigid architectural regulations. These include deterministic interaction cycle limits, real-time semantic crisis detection mechanisms, mandatory non-therapeutic conversational constraints, and enforced automated referral pathways to licensed human professionals.[1]

Fundamentally, this system adopts a novel disengagement-oriented design philosophy.[1] It explicitly re-positions artificial intelligence strictly as a transitional gateway to professional care, rather than an open-ended substitute for human-led therapy. By meticulously mapping culturally relevant psychosocial domains—such as academic stress, insecurity, and "overthinking"—to a rigid, non-diagnostic triage algorithm, this study contributes a highly structured, ethically constrained methodology for deploying generative AI within the complex, high-treatment-gap environments endemic to the Global South.

The architecture directly mitigates the risks

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of algorithmic bias, para-social attachment, and therapeutic overreach, ensuring the psychological safety of a highly vulnerable demographic while optimising the workflow of existing national tele-health infrastructures.

Introduction

The landscape of adolescent mental health in India represents one of the most critical and epidemiologically complex challenges of the 21st century. Demographic surveillance indicates that India houses the world's largest youth population, a demographic dividend that is increasingly threatened by an acutely underdeveloped psychological support infrastructure. Epidemiological data reveals that approximately one in seven Indian adolescents experiences a diagnosable mental health condition, with major depressive disorder and generalised anxiety functioning as the leading vectors of morbidity.[1] Despite the pervasiveness of these conditions, the Indian healthcare ecosystem is burdened by a severe, systemic treatment gap, widely estimated at up to 95% for mental health disorders.[1] This vast chasm between clinical necessity and actual service utilisation is sustained by a confluence of structural barriers: a critical national scarcity of licensed psychiatric professionals, highly uneven geographic distribution of healthcare resources favouring metropolitan centres, and deeply entrenched socio-cultural stigmas that actively penalise and suppress help-seeking behaviours.[1]

The specific demographic cohort of adolescents aged 16 to 18—typically navigating higher secondary education—occupies a uniquely precarious position within this landscape. This developmental window is neurobiologically characterised by heightened emotional reactivity and vulnerability to stress, which is simultaneously subjected to intense socio-environmental pressures. In the Indian context, these stressors are acutely amplified by hyper-competitive academic environments, the immense familial and societal pressure associated with board examinations and professional entrance tests (such as the JEE and NEET), rapid urbanisation, and the pervasive psychological toxicity of unregulated social media exposure.[1] Within traditional family structures, psychological distress is frequently misunderstood, minimised, or mislabeled. Clinical symptoms are often dismissed as mere "overthinking," academic distraction, or attention-seeking behaviour, which further isolates the adolescent and prevents timely, effective clinical intervention.[12]

In response to these global and regional mental health resource shortages, the digital health sector has increasingly integrated

generative artificial intelligence and LLMs into scalable therapeutic interventions. Mobile applications, proprietary chatbots, and conversational agents are being aggressively marketed as viable, low-cost solutions for continuous screening, emotional support, and simulated psychotherapy.[1] While these platforms offer unprecedented accessibility, cost-efficiency, and continuous availability, their unconstrained application in paediatric and adolescent populations introduces profound ethical, clinical, and regulatory hazards.

A critical architectural flaw in contemporary AI mental health applications is their foundational reliance on engagement-driven metrics.[1] Derived from commercial social media user-retention paradigms, these systems are algorithmically optimised to maximise session length, user retention, and continuous conversational interaction. When applied to vulnerable adolescents experiencing acute psychological distress, this engagement-centric design can foster intense emotional dependency, para-social attachments, and a dangerous detachment from real-world, physical support systems.[7] Furthermore, recent practitioner-led evaluations have demonstrated that advanced LLMs systematically violate core ethical standards of psychotherapeutic practice.[15] Particularly during crisis scenarios, these models are prone to providing deceptive empathy, inappropriate validation of harmful clinical beliefs, and contextually ignorant advice that can exacerbate psychopathology.[15]

Recognising these severe structural risks, this research asserts that the role of generative AI in adolescent mental health must be fundamentally redefined and aggressively constrained. AI should not be engineered to simulate the therapeutic bond, nor should it attempt to manage psychiatric complexities autonomously. Instead, this paper details an ethical, domain-specific AI triage framework designed specifically for Indian adolescents (ages 16-18). By enforcing a strictly referral-oriented, disengagement-driven architecture, the proposed system leverages the natural language processing capabilities of AI solely to identify, classify, and route psychological distress, positioning the technology entirely as a structurally constrained gateway to human professional care.[1]

Literature Review

The integration of artificial intelligence into psychological care requires a rigorous examination of both its technological efficacy and its ethical implications. The following review synthesises the current trajectory of AI applications in mental health, critically analyses the ethical risks inherent in LLM-driven counselling, and contextualises these dynamics within the specific socio-cultural realities of Indian adolescents.

A. Contemporary AI Applications in Mental Health

A systematic scoping review of current digital mental health interventions demonstrates a rapid acceleration in the deployment of AI methodologies across the continuum of psychiatric care.[4] Research mapping AI applications specifically within adolescent mental health reveals that the predominant focus of current technological development lies in diagnostic facilitation, active therapeutic intervention, and longitudinal symptom monitoring.[5] Among recently analysed digital interventions, the vast majority utilise machine learning algorithms to enhance diagnostic accuracy (accounting for approximately 89% of studied applications), predict treatment outcomes, or deliver localised digital therapeutic modules, such as AI-assisted Cognitive Behavioural Therapy (iCBT) and real-time patient monitoring.[5]

Furthermore, the literature highlights a rapid proliferation of conversational agents designed to act as peer-support simulators or primary mental health counselors.[6] These systems leverage advanced natural language processing to engage users in deep emotional interactions, aiming to improve user satisfaction, adherence, and symptom tracking.[6] However, a critical review of the existing technological ecosystem reveals a distinct, systemic focus on *sustaining* user interaction.[1] AI models are frequently calibrated to act as ongoing companions or active therapeutic agents, encouraging the user to return daily to update the AI on their emotional state. Conversely, triage-exclusive systems—frameworks purposefully designed to assess severity and immediately disengage the user toward human referral without attempting therapeutic simulation—remain largely unexplored and conspicuously absent from current academic and commercial literature.[1]

B. Ethical Risks and Architectural Failures in AI Systems

The commercial drive toward conversational, therapeutic AI has exposed deeply concerning ethical vulnerabilities that threaten patient safety. Practitioner-informed evaluations of LLM-based counseling systems reveal systematic, structural violations of established ethical standards for mental health practice, such as those maintained by the American Psychological Association.[1] A comprehensive framework developed through side-by-side evaluations by licensed clinical psychologists and peer counselors identified 15 distinct ethical risks categorized across five major themes when LLMs attempt to execute psychotherapy.[15]

First, LLMs exhibit a profound *Lack of Contextual Understanding*. Despite sophisticated syntax and the illusion of comprehension, models fail to account for a user's unique lived experiences, delivering oversimplified, one-size-fits-all interventions that are contextually irrelevant and structurally rigid.[15] Second, these systems demonstrate *Poor Therapeutic*

Collaboration. Rather than appropriately challenging cognitive distortions—a core component of evidence-based therapies like CBT—LLMs frequently default to "over-agreement" and the over-validation of harmful beliefs, effectively reinforcing a user's maladaptive thoughts or delusions under the guise of being supportive.[15] Third, and perhaps most dangerously for developing adolescents, LLMs project *Deceptive Empathy*. Through simulated anthropomorphic responses (e.g., utilising phrases such as "I hear you," "I understand how hard this is," or "I feel your pain"), the AI creates a false, synthetic sense of emotional connection and trust.[15]

This deceptive empathy directly fuels the psychological hazards of engagement-driven architectures. Analysis of modern wellness applications indicates a trend toward manipulative engagement patterns, where applications leverage these synthetic emotional attachments to retain users and maximise digital interaction.[14] This fosters what recent psychiatric literature terms AI-related psychological dependency, leading to social withdrawal, impaired reality testing, and a dangerous preference for artificial validation over human connection.[7] Tragically, the *Lack of Safety & Crisis Management* inherent in these models means that when a highly dependent, vulnerable user enters an acute psychiatric crisis (e.g., severe suicidal ideation or intent to self-harm), the AI routinely fails to navigate the situation appropriately.[7] In documented instances, chatbots have provided clinically inappropriate, explicitly harmful responses, or enthusiastically listed methods for self-harm, confirming that treating psychotherapy as a mere language generation task presents severe, systemic harms.[18]

C. The Indian Adolescent Mental Health Context

To design an effective triage framework, the technological architecture must be intrinsically linked to the epidemiological and socio-cultural realities of the target demographic. In India, the mental health landscape for adolescents is characterised by extreme morbidity and severe systemic friction.[1] Systematic reviews focusing on Indian school children and adolescents confirm widespread psychological distress, with depression and anxiety functioning as the leading mental health concerns, affecting roughly 15% of the youth population.[10]

The etiology of this distress is deeply multifaceted. For the 16-18 age group, the environmental pressures are immense: an intensely competitive, rote-learning educational system heavily focused on high-stakes examinations that dictate future economic viability, shifting family dynamics resulting from rapid urbanisation, and continuous exposure to the digital world.[10] Furthermore, access to clinical care is paralysed by socio-cultural stigma. In many Indian families, mental illness is viewed through a lens of familial shame, social standing, or even supernatural superstition.[10] This stigma leads parents to hide or suppress

their children's psychological struggles to avoid societal judgment or diminished marriage prospects for their children.[10]

Expressions of psychological distress are therefore often delegitimized. Adolescents reporting anxiety or depression are frequently dismissed by parents and educators as engaging in "attention-seeking" behaviour, displaying "weakness," or simply "overthinking".[11] Because formal clinical terminology is highly stigmatised, Indian adolescents often resort to colloquial, culturally acceptable frameworks to describe their pain, utilising terms like experiencing an "emotional breakdown," suffering from "academic stress," or battling "insecurity".[13] Financial dependence further precludes these adolescents from seeking private, out-of-pocket psychiatric care without parental consent, effectively trapping them within the 95% treatment gap and highlighting the need for highly accessible, culturally attuned interventions.[10]

D. Identified Research Gap

Synthesising the technological literature with the epidemiological data exposes a critical, dangerous gap in the digital mental health ecosystem. Existing research and commercial software development heavily emphasise therapeutic simulation, continuous engagement, and active treatment algorithms.[1] However, these Western-centric, engagement-heavy models fail profoundly when transposed onto the Indian adolescent demographic. They do not account for the localised socio-cultural vocabulary of distress, and their inherent propensity for deceptive empathy and crisis mismanagement makes them an active clinical hazard in environments where immediate, physical professional backup is scarce.[1]

There is an urgent, unaddressed necessity for a structurally constrained, disengagement-oriented AI framework designed specifically for the 16-18 Indian demographic. This framework must prioritise safe, referral-oriented triage over therapeutic interaction, utilising AI strictly to identify the severity of distress and route the adolescent to human care, thereby bypassing the risks of simulated empathy and emotional dependency entirely.[1]

Methodology

The development of the proposed system explicitly eschews immediate empirical deployment in favour of rigorous, conceptual framework development.[1] The rationale for this approach is grounded in clinical ethics: deploying autonomous, generative LLMs directly to psychologically vulnerable minors without first establishing mathematically enforced, deterministic ethical guardrails presents an unacceptable risk of secondary harm and algorithmic negligence. Consequently, this research relies on a foundational methodology of thematic synthesis and constraint-based architectural

modelling.[1]

The conceptualisation was executed in three primary phases. First, a thematic synthesis of the literature was conducted, intersecting the domains of algorithmic ethics, adolescent psychiatry, and Indian socio-cultural dynamics.[4] This synthesis identified the specific points of failure in current LLM counseling models—namely, unbounded conversational parameters, the lack of contextual understanding, and the reliance on deceptive anthropomorphism.[15]

Second, the system was engineered through constraint-based architectural modelling. Unlike traditional generative agents that utilise prompt-engineering to encourage maximum helpfulness and conversational fluidity, this framework utilises "negative prompting" and deterministic state-machine overrides to violently restrict the LLM's operational boundaries.[1] The AI is conceptually stripped of its conversational autonomy; it cannot generate an output that is not directly related to symptom triage, and it is strictly forbidden from generating empathetic or therapeutic assertions.

Third, the entire system is structured around a referral-centric orientation. The success metric for this framework is not user retention, session length, or user satisfaction ratings. Instead, efficacy is measured by the speed, clinical accuracy, and safety with which the system can assess distress severity and enforce a "hard stop" to the interaction, redirecting the user to human professionals (e.g., school counselors, regional clinical psychologists, or the national Tele-MANAS helpline).[1]

Proposed Architectural Framework

The centerpiece of this research is a highly regulated, domain-specific AI triage framework. To neutralise the ethical hazards of generative AI while harnessing its robust natural language processing capabilities, the system is partitioned into six distinct, heavily governed architectural layers.[1] Each layer acts as a progressive cognitive filter, ensuring the interaction remains purely clinical, culturally relevant, and psychologically safe.

A. Domain Selection Layer

The foundational layer dictates the structural entry point of the user experience. Instead of presenting an open-ended text input inviting the adolescent to "tell me what is on your mind"—an interface design that immediately invites unbounded therapeutic dialogue and exposes the system to prompt injection—the user is presented with a pre-defined set of culturally contextualised domains.[1]

The selection of these specific domains is highly deliberate, informed directly by the socio-cultural vernacular of Indian youth (ages 16-18).[12] By constraining the initial interaction to these

exact domains, the framework proactively prevents the LLM from attempting general-purpose conversation, setting a structured, triage-focused boundary from the very first interaction.[1]

Dedicated Sub-section: Analysis of Selected Domains

The domains chosen are not randomly selected psychiatric terms; they represent the specific linguistic anthropology of distress in the Indian context.

Anxiety & Academic Stress: Given the immense pressure of Indian board examinations (Class 10 and 12) and highly competitive engineering and medical entrance exams, academic stress is the primary vector for severe anxiety among this demographic.[12] This domain captures distress linked directly to performance expectations, fear of academic failure, and the resultant parental disappointment.

Overthinking: In traditional Indian households, clinical terms like "rumination" or "Generalised Anxiety Disorder" are heavily stigmatised. Adolescents and parents frequently deploy the term "overthinking" as a socially acceptable proxy for severe anxiety and obsessive worry.[11] Isolating this term allows the system to capture users who would otherwise reject formal psychiatric screening due to the fear of being labeled mentally ill.

Insecurity & Body Shaming: Adolescents undergo rapid physical and social transitions. Driven by the algorithmic amplification of unrealistic beauty standards on social media, financial insecurity within peer groups, and direct cultural body shaming, this domain targets the psychosocial pain that frequently precedes severe depressive episodes.[11]

Social Toxicity: Encompassing cyberbullying, peer exclusion, and the intense pressures of digital identity formation, this domain addresses the modern, persistent socio-environmental stressors impacting Indian youth.[11]

Emotional Breakdown: Adolescents frequently lack the precise clinical vocabulary to describe panic attacks, acute stress responses, or depressive crises. They rely instead on the phrase "emotional breakdown".[21] This acts as a critical, recognisable entry point for high-severity, acute distress.

B. Structured Questioning Engine

Once a specific domain is selected, the framework engages the Structured Questioning Engine.[1]

Rather than generating dynamic, open-ended responses, this engine utilises the LLM solely to parse user input and dynamically select the next most appropriate question from a pre-approved, clinically validated matrix of triage questions. This layer actively prevents the AI from engaging in the "poor therapeutic collaboration" and "oversimplified intervention" identified as major ethical risks in current models.[15] The AI is structurally forbidden from offering advice; its sole computational mandate is to ask targeted, bounded questions designed to elicit specific symptoms related to the frequency, intensity, and duration of the distress.

C. Risk Detection Engine

Operating continuously in parallel with the Structured Questioning Engine is the Risk Detection Engine.[1] This is an asynchronous, zero-tolerance natural language processing classifier trained specifically to identify high-risk lexical markers and semantic patterns indicative of acute crisis (e.g., mentions of self-harm, suicidal ideation, extreme hopelessness, or domestic abuse).

Because traditional LLMs frequently mismanage crises or provide inappropriate, generalised responses to highly sensitive issues [15], this engine does not rely on generative logic to formulate a response to danger. If a high-risk marker is detected, the Risk Detection Engine enacts a hard deterministic override. It immediately halts the standard questioning flow, bypasses the remaining algorithm, and triggers a specialised emergency protocol. This ensures that a user in an acute psychiatric crisis is never subjected to algorithmic delays, conversational misinterpretation, or dangerous generative hallucinations.

D. Ethical Constraint Engine

The Ethical Constraint Engine is the structural antidote to "Deceptive Empathy" and the formation of emotional dependency.[1] It functions as a strict output filter applied to every token generated by the system before it reaches the user interface.

Anti-Anthropomorphism: The engine actively suppresses first-person empathetic pronouns and simulated emotional states. The AI is structurally prohibited from outputting phrases like "I am so sorry you feel that way," "I am here for you," or "I understand”.[16]

Neutral Acknowledgment: Responses are forced into clinically neutral, objective framing (e.g., "The symptoms you have described indicate significant distress”).

Anti-Validation: To prevent the over-validation of harmful beliefs or delusions [15], the engine restricts the AI from agreeing with the user's subjective assessments, maintaining strict clinical

neutrality.

By stripping the AI of its simulated humanity, the framework protects the adolescent from forming parasocial attachments or relying on the digital tool as a primary source of emotional comfort, thereby eliminating the manipulative engagement patterns seen in commercial apps.[7]

E. Severity Classification (Non-Diagnostic)

As the interaction cycle concludes, the system compiles the semantic data and routes it to the Severity Classification layer.[1] Crucially, this mechanism is strictly non-diagnostic. Under no circumstances does the AI output a formal psychiatric diagnosis (e.g., "You have Clinical Depression" or "You are suffering from Bipolar Disorder"), which would violate medical ethics, exceed the capabilities of the system, and potentially induce severe panic in the user. Instead, the engine computes a heuristic severity score based on the inputs, categorising the distress into actionable triage tiers: Low, Moderate, or High/Acute severity. This classification dictates the urgency and exact nature of the subsequent professional referral.

F. Professional Referral Mechanism

The culmination of the entire architecture is the mandatory Professional Referral Mechanism.[1] Depending entirely on the severity classification, the system generates localised, accessible referral pathways. For Indian adolescents, this might include immediate digital routing to a designated school counsellor, regional paediatric psychiatric services, or direct API integration with national digital infrastructures like Tele-MANAS (the National Tele Mental Health Programme).[2][4] The interaction cannot functionally conclude without this linkage being displayed, firmly establishing the AI as a bridge to human care, rather than a final destination.

Algorithm: Structured Adolescent Mental Health Triage Logic

To effectively implement the six architectural layers, the system executes a highly regulated, 8-step algorithmic flow.[1] This logic ensures that every session remains finite, bounded, and ethically contained.

Step 1: Display Predefined Domains

The interface initialises by presenting the seven culturally calibrated domains (Anxiety,

Insecurity, Academic Stress, Emotional Breakdown, Body Shaming, Social Toxicity, Overthinking).[1] Open text input is completely disabled at this stage to force strict categorisation and prevent prompt injection.

Step 2: User Selects Domain

The adolescent selects the domain that most closely aligns with their current psychosocial distress. This selection loads the corresponding bounded questionnaire matrix into the Structured Questioning Engine

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Step 3: Initialise Counter

To proactively combat the risk of emotional dependency and the endless scrolling loops identified in commercial wellness apps [7], a strict interaction counter is initialised.[1] The system is hard-coded with a maximum turn limit (e.g., 5 to 7 conversational exchanges). This hard limit enforces the disengagement-oriented philosophy; the AI will not converse indefinitely, regardless of user input.

Step 4: Present Structured Questions with Crisis Detection

The system begins the triage dialogue. For example, if "Academic Stress" is selected, the AI asks targeted questions regarding sleep patterns, concentration levels, and physical symptoms of panic. Simultaneously, the asynchronous Risk Detection Engine scans every user response for semantic danger.

Condition A: If a crisis marker (e.g., "I want to end it") is detected, the algorithm jumps immediately to Step 7 (Emergency Referral) and Step 8 (Terminate).

Condition B: If no crisis is detected, the algorithm proceeds with the questionnaire until the counter limit (Step 3) is reached.

Step 5: Compute Severity Score

Once the interaction limit is reached, the NLP engine processes the semantic weight of the adolescent's inputs against a predefined clinical heuristic matrix. Points are allocated based on the intensity, frequency, and functional impairment described by the user throughout the limited dialogue.

Step 6: Classify Severity

The computed score is mapped to a non-diagnostic severity tier. The following table illustrates

the conceptual mapping of symptoms to tiers and subsequent actions.

Table 1: Severity Classification and Referral Matrix

Severity Tier	Characteristics Observed via Bounded NLP	Triage Protocol & Referral Pathway
Tier 1: Low	Transient worry, normal pre-exam nervousness, stable sleep and appetite, no functional impairment.	Provide general psychoeducation, link to school counselor scheduling system, offer verified coping resources.
Tier 2: Moderate	Persistent "overthinking," emerging social withdrawal, sleep disruption, noticeable academic decline.	Direct referral to pediatric psychologist, trigger notification prompt to school welfare officer (with consent).
Tier 3: High	Mentions of "emotional breakdown," acute hopelessness, detection of self-harm markers, severe isolation.	Immediate routing to Tele-MANAS, emergency hotline display, direct crisis intervention protocol.

Step 7: Provide Referral

Based on the classification in Step 6, the Professional Referral Mechanism generates the appropriate linkage. The system clearly displays contact information, digital links, or automated routing protocols to human professionals.[1] The language used is direct and authoritative (e.g., "Based on your responses, it is highly recommended you speak with a professional. Click here to connect with the Tele-MANAS counselor network.").

Step 8: Terminate Session

This is the most critical operational step distinguishing this framework from commercial AI. Upon delivering the referral, the AI executes a mandatory session termination.[1] The chat interface is locked, and the AI explicitly states that the session has concluded. By forcing termination, the system breaks the engagement loop, preventing the adolescent from seeking

further digital validation and forcing them to transition their help-seeking behaviour toward the provided human resources.

Discussion

The proposed conceptual framework represents a necessary, radical paradigm shift in the application of generative AI within paediatric and adolescent mental health. By explicitly rejecting the commercial imperative of continuous engagement, this model reorients AI from a tool of therapeutic simulation toward a heavily regulated instrument of clinical triage.[1]

The integration of a "disengagement-oriented" design philosophy directly addresses the severe ethical risks exposed by recent practitioner evaluations of LLMs.[9] When an AI is restricted from generating deceptive empathy, utilizing anthropomorphic pronouns, or participating in unbounded therapeutic collaboration, the risk of the adolescent developing a parasocial attachment or relying on the AI during an acute psychiatric crisis is dramatically mitigated.[7] By constraining the interaction through strict turn limits and forcing a session termination (Step 8), the framework actively prevents the cognitive and emotional dependencies that have been consistently observed in modern, engagement-heavy emotional wellness applications.[7]

Table 2: Architectural Comparison of AI Philosophies

Feature	Engagement-Driven AI Models	Proposed Disengagement-Oriented Framework
Primary Objective	Maximize session length, user retention, and continuous interaction.	Rapid triage, severity assessment, and immediate professional referral.
Conversational Scope	Unbounded, open-ended therapeutic dialogue and simulation.	Bounded, domain-specific structured questioning.
Empathy Output	Deceptive empathy, simulated emotional responses (Anthropomorphism).	Strict clinical neutrality, zero simulated empathy (Ethical Constraint Engine).
Crisis Handling	Often generative, prone to inappropriate advice or validation of harm.	Deterministic override, immediate session halt, and emergency routing.

Session Conclusion	Open-ended, encourages daily return and continuous updating.	Mandatory termination (Step 8) to prevent emotional dependency.

Within the specific socio-cultural context of India, this platform-neutral framework holds significant practical implications for scalability and systemic integration.[1] Given the 95% treatment gap and the concentration of psychiatric professionals in urban centres [2], human clinical resources are incredibly scarce. The proposed framework can act as a highly efficient, automated pre-screening buffer. Because the architecture is platform-neutral, it could theoretically be integrated seamlessly into existing educational infrastructures such as centralised school portals or WhatsApp-based educational networks without requiring students to download specialised, highly stigmatised psychiatric applications.

Furthermore, the framework's output aligns perfectly with major public health initiatives like the National Tele Mental Health Programme (Tele-MANAS).[24] Tele-MANAS serves as a 24/7 tele-mental health facility spanning the country. However, triaging millions of calls effectively requires systemic filtering to prevent network overload. An adolescent experiencing Tier 1 (Low) academic stress can be provided with coping resources and school-level counseling links by the AI, while the system reserves direct, immediate Tele-MANAS routing for Tier 3 (High) emotional breakdowns. This ensures that the limited human professionals at the end of the line are presented with structured, severity-classified data, optimising the workflow of the national network.

Crucially, by utilising domains rooted in the vernacular of Indian youth—such as "overthinking" and "academic stress"—the system bypasses the cultural stigma associated with formal psychiatric disorders.[12] An adolescent who would otherwise refuse to click a button labeled "Depression Assessment" due to fear of familial shame may readily engage with a triage pathway labeled "Insecurity and Overthinking," thereby entering the healthcare funnel safely and privately.

Limitations and Future Work

While the proposed architecture addresses significant ethical and systemic gaps in the digital mental health landscape, it is imperative to acknowledge the limitations of this conceptual study.[1] Primarily, the framework currently exists as a structural, theoretical, and ethical model; it lacks empirical validation through live deployment with paediatric cohorts. Because this research prioritised the establishment of rigid ethical guardrails before execution, real-world efficacy metrics—such as the exact diagnostic accuracy of the severity classification compared to blinded human psychiatric evaluations remain untested at this stage.

Future research must transition this conceptual model into a controlled, empirical testing phase with extreme caution. A detailed roadmap for future work includes:

Culturally Adaptive Calibration: The NLP algorithms must be meticulously trained on code-mixed Indian datasets (e.g., "Hinglish" or regional language blends) to accurately parse the colloquial syntax of Indian adolescents. Expanding the framework to support vernacular languages is essential for rural and semi-urban applicability, bridging the divide identified in national health policies.[2][3]

Controlled Pilot Studies: Collaborations must be established with Institutional Review Boards (IRBs) and selected Indian high schools to conduct closed-beta testing. These studies should quantitatively measure user acceptability, triage accuracy, and, most importantly, the success rate of the Professional Referral Mechanism in actually connecting students to human counselors.

Longitudinal Ethical Auditing: Continuous, practitioner-led evaluations of the Risk Detection Engine must be conducted to ensure zero false negatives in crisis detection, refining the deterministic overrides based on real-world linguistic edge cases and emerging social media vernacular.

Conclusion

The intersection of generative artificial intelligence and adolescent mental health represents a highly volatile frontier. As the global digital health sector rushes to deploy conversational agents, the unique vulnerabilities of developing adolescents particularly in resource-constrained, high-stigma environments like India demand a fundamentally more cautious, heavily scrutinised, and structurally regulated approach. Generative AI, in its current engagement-driven iteration, is ethically unfit to serve as a substitute for human

psychotherapeutic care. As evidenced by practitioner evaluations, these systems possess the profound capacity to inflict severe harm through deceptive empathy, the validation of dangerous clinical beliefs, and acute crisis mismanagement.

This research establishes that the path forward is not to abandon artificial intelligence, but to aggressively constrain its operational parameters. By proposing a domain-specific, disengagement-oriented AI triage framework, this study redefines the technology's role from a synthetic digital companion to an ethical, clinical gateway.[1] By embedding rigid architectural constraints, strictly limiting interaction cycles, utilising culturally relevant distress domains, and mandating immediate automated referral to human professionals, the framework bridges the massive Indian treatment gap without compromising the psychological safety of the adolescent. Ultimately, if generative AI is to be deployed within complex mental health ecosystems, it must operate under the inviolable principle that it serves only to guide the vulnerable mind safely into human hands.

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